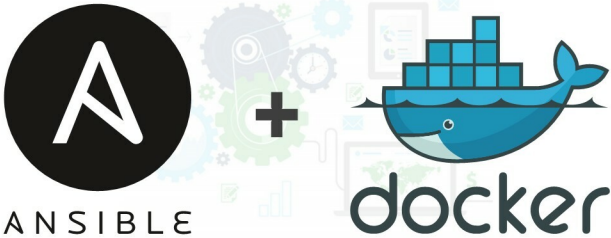


DevOps Series:

Provisioning with Ansible

Ansible is the simplest way to automate apps and IT infrastructure. It meshes well with Docker to deploy apps. In this ninth article in the series on DevOps, we explore the use of Ansible for launching Docker containers and provisioning virtual machines.



Provisioning is the first step in an application's deployment process. In a cloud environment, software can be run from a Docker container, virtual machine or bare metal, and Ansible can be used for provisioning such systems. In this article, we explore how to use Ansible to launch Docker containers and provision virtual machines.

Setting it up

Let's create an Ansible playbook for the 'Get started with Docker Compose' `composetest` example available at <https://docs.docker.com/compose/gettingstarted/>. The Ansible version used on the host system (Ubuntu x86_64) is 2.2.0.0. You will need to install Docker CE and `docker-compose` on Ubuntu. Follow the installation guide provided at <https://docs.docker.com/engine/installation/linux/docker-ce/ubuntu/#install-using-the-repository> to install Docker CE. You can then install `docker-compose` using the APT package manager:

```
$ sudo apt-get install docker-compose
```

The `composetest/` folder consists of the following files:

```
composetest/app.py
composetest/docker-compose.yml
composetest/Dockerfile
composetest/provision.yml
```

`composetest/requirements.txt`

The `app.py` file contains a basic Flask application that communicates with a backend Redis database server. Its file contents are as follows:

```
from flask import Flask
from redis import Redis

app = Flask(__name__)
redis = Redis(host='redis', port=6379)

@app.route('/')
def hello():
    count = redis.incr('hits')
    return 'Hello World! I have been seen {} times.\n'.
format(count)

if __name__ == "__main__":
    app.run(host="0.0.0.0", debug=True)
```

An HTTP request to the Flask application returns the text string 'Hello World! I have been seen N times.' This will be run inside a Docker container. The `requirements.txt` file is provided to list the dependencies required for the project:

```
flask
```